

Gaussian Process II

$$x_1, \dots, x_n \stackrel{i.i.d.}{\sim} N(0, \sigma^2), \quad \bar{x} = \frac{1}{n} \sum_{k=1}^n x_k, \quad \bar{s} = \frac{1}{n-1} \sum_{k=1}^n (x_k - \bar{x})^2$$

Sample Mean

Sample Variance

有: $E(\bar{x}) = \mu, E(\bar{s}) = \sigma^2$

~ 无偏估计

Thm: $\bar{x}$ is independent of $\bar{s}$

Pf:
$$\begin{aligned} \sum_{k=1}^n (x_k - \bar{x})^2 &= \sum_{k=1}^n x_k^2 - 2 \left(\sum_{k=1}^n x_k \right) \bar{x} + (\bar{x})^2 n \\ &= \sum_{k=1}^n x_k^2 - n \bar{x}^2 \\ &= \sum_{k=1}^n x_k^2 - \left(\frac{1}{\sqrt{n}} \sum_{k=1}^n x_k \right)^2 \end{aligned}$$

作变换 $A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$, 满足 $\sum_{k=1}^n x_k^2 = \sum_{k=1}^n y_k^2$, $\sum_{k=1}^n \frac{1}{\sqrt{n}} x_k = y_1$,
A 为正交矩阵

取 $A = \begin{pmatrix} \frac{1}{\sqrt{n}} & \cdots & \frac{1}{\sqrt{n}} \\ & * & \end{pmatrix}$, 只需满足形为此的正交阵即可

$x = (x_1, \dots, x_n)^T$, 取 $x \sim N(\mu \cdot 1, \sigma^2 I) \Rightarrow y \sim N(A \cdot \mu \cdot 1, \sigma^2 I)$

因此 $y_1 = \frac{1}{\sqrt{n}} \sum_{k=1}^n x_k$, $(n-1)\bar{s} = \sum_{k=1}^n x_k^2 - \left(\frac{1}{\sqrt{n}} \sum_{k=1}^n x_k \right)^2 = \sum_{k=1}^n y_k^2 - y_1^2 = \sum_{k=2}^n y_k^2$

从而 $\bar{x}$ 独立于 $\bar{s}$. □

Conditional Distribution of Gaussian

$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}^T \in \mathbb{R}^{m \times (n_1 + n_2)}$, $x \sim N(\mu, \Sigma)$

→ $x_1 | x_2$ 服从什么分布? Gaussian

$$\begin{aligned} f_{x_1|x_2}(x_1|x_2) &= \frac{f_{x_1, x_2}(x_1, x_2)}{f_{x_2}(x_2)}, \quad \text{作分块 } \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \\ &= C \frac{\exp\left(-\frac{1}{2} \begin{pmatrix} x_1^T - \mu_1^T & x_2^T - \mu_2^T \end{pmatrix} \Sigma^{-1} \begin{pmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \end{pmatrix}\right)}{\exp\left(-\frac{1}{2} (x_2^T - \mu_2^T) \Sigma_{22}^{-1} (x_2 - \mu_2)\right)} \quad \dots \textcircled{1} \end{aligned}$$

与 Diffusion 是类似的, 进行配方. 为此对 Σ 进行分块对角化

考虑 $\begin{pmatrix} I & -\Sigma_{12} \Sigma_{22}^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{pmatrix} I & 0 \\ -\Sigma_{22}^{-1} \Sigma_{21} & I \end{pmatrix} = \begin{pmatrix} \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} & 0 \\ 0 & \boxed{\Sigma_{22}} \end{pmatrix}$
留着抵消

取 $\begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}^{-1} = \begin{pmatrix} I & 0 \\ -\Sigma_{22}^{-1} \Sigma_{21} & I \end{pmatrix} \begin{pmatrix} (\Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21})^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} \begin{pmatrix} I & -\Sigma_{12} \Sigma_{22}^{-1} \\ 0 & I \end{pmatrix}$

$$\begin{pmatrix} x_1^T - \mu_1^T & x_2^T - \mu_2^T \end{pmatrix} \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}^{-1} \begin{pmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \end{pmatrix} \\ = \begin{pmatrix} x_1^T - \mu_1^T - (x_2^T - \mu_2^T) \Sigma_{22}^{-1} \Sigma_{21} & x_2^T - \mu_2^T \end{pmatrix} \begin{pmatrix} (\Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21})^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} \begin{pmatrix} x_1 - \mu_1 - \Sigma_{12} \Sigma_{22}^{-1} (x_2 - \mu_2) \\ x_2 - \mu_2 \end{pmatrix}$$

恰好为方程

$$= \begin{pmatrix} x_1^T - \mu_1^T - (x_2^T - \mu_2^T) \Sigma_{22}^{-1} \Sigma_{21} \end{pmatrix} (\Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21})^{-1} \begin{pmatrix} x_1 - \mu_1 - \Sigma_{12} \Sigma_{22}^{-1} (x_2 - \mu_2) \end{pmatrix} + \underbrace{(x_2^T - \mu_2^T) \Sigma_{22}^{-1} (x_2 - \mu_2)}_{\text{恰好为下面的分母}}$$

回到①式，故：

$$E(x_1 | x_2) = \mu_1 + \underbrace{\Sigma_{12} \Sigma_{22}^{-1} (x_2 - \mu_2)}_{\text{一维视角}} \quad \Sigma_{x_1|x_2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \quad \begin{matrix} \text{半正定的} \\ \sim \sigma_{11} - \frac{\sigma_{12} \sigma_{21}}{\sigma_{22}} \geq 0 \\ \rightarrow \text{Cauchy 不等式} \end{matrix}$$

投影 Projection

→ 将 x_2 中的信息吸收到 x_1 中。

注： $E(x_1 | x_2)$ 为均方意义下的最优估计
Projection 为均方意义下的最优线性估计

在高斯意义下，Optimal = Optimal Linear

Linear Gaussian System.

很为系统存在内在状态，我们只能得到观测数据。

Model: $Y = AX + \varepsilon$, $X \sim N(\mu, \Sigma_x)$, $\varepsilon \sim N(0, \Sigma_N)$, X 独立于 ε

我们来研究 $X|Y$.

首先说明 (X, Y) 为联合高斯。

pf:
$$\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} I & 0 \\ A & I \end{pmatrix} \begin{pmatrix} X \\ \varepsilon \end{pmatrix}$$

由于 $X \sim N$, $\varepsilon \sim N$, 且 X, ε 相互独立。故 (X, ε) 为联合高斯。

而 $\begin{pmatrix} I & 0 \\ A & I \end{pmatrix}$ 为一个线性变换。

因此 (X, Y) 为联合高斯。

□

对于 $X|Y$, $E(X) = \mu$, $\Sigma_{XY} = E(X - \mu_X)(Y - \mu_Y)^T$

$$\begin{aligned} &= E(X - \mu_X)(AX + \varepsilon - A\mu_X)^T \\ &= E(X - \mu_X)(A(X - \mu_X) + \varepsilon)^T \\ &= \Sigma_X A^T \end{aligned}$$

$$\begin{aligned} \Sigma_{YY} &= E(Y - \mu_Y)(Y - \mu_Y)^T = E(A(X - \mu_X) + \varepsilon)(A(X - \mu_X) + \varepsilon)^T \\ &= A\Sigma_X A^T + \Sigma_N \end{aligned}$$

故 $E(X|Y) = \mu + \Sigma_X A^T (A\Sigma_X A^T + \Sigma_N)^{-1} (Y - A\mu_X)$

$$\Sigma_{X|Y} = \Sigma_X - \Sigma_X A^T (A\Sigma_X A^T + \Sigma_N)^{-1} A\Sigma_X$$

eg. $X, Y \stackrel{i.i.d}{\sim} N(0, 1)$, 计算 $E(X+Y | X-Y)$

$$E(X+Y | X-Y) = E(X+Y) + E((X+Y - \mu_{X+Y})(X-Y - \mu_{X-Y})) (\text{Var}(X-Y))^{-1} (X-Y - \mu_{X-Y}) \\ = 0 + 0 = 0$$

另一方面, 也可以考虑 $\begin{pmatrix} X+Y \\ X-Y \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}$

由于 $\begin{pmatrix} X \\ Y \end{pmatrix} \sim N(0, I)$, 故 $\begin{pmatrix} X+Y \\ X-Y \end{pmatrix} \sim N\left(\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot 0, \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} I \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\right)$
 $= N(0, 2I)$

故 $E(X+Y | X-Y) = E(X+Y) = 0$ 

eg. 进一步计算 $E((X+Y)^2 | X-Y) = E(X+Y)^2 = \text{Var}(X+Y) = 2$

$$E((X+2Y)^2 | X-Y)$$

$$\begin{pmatrix} X+2Y \\ X-Y \end{pmatrix} \sim N\left(0, \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 5 & -1 \\ -1 & 2 \end{pmatrix}\right)$$

故 $X+2Y$ 与 $X-Y$ 不独立

① $Z = X+2Y | X-Y \sim N(\mu_{X+2Y|X-Y}, \sigma_{X+2Y|X-Y}^2)$

② $E(g(X+2Y) | X-Y) = E(g(Z))$

$$Z \sim N\left(0 + (-1) 2^{-1} (X-Y - 0), 5 - (-1) 2^{-1} (-1)\right) \\ = N\left(-\frac{1}{2}(X-Y), \frac{9}{2}\right)$$

$$E(Z^2) = (E(Z))^2 + \text{Var}(Z) = \frac{1}{4}(X-Y)^2 + \frac{9}{2}$$